Computer-Aided Civil and Infrastructure Engineering 49 (2026) 100044

Contents lists available at ScienceDirect

Computer-Aided Civil and Infrastructure Engineering

journal homepage: www.sciencedirect.com/journal/computer-aided-civil-and-infrastructure-engineering

Research article

AI-driven human-in-the-loop structural optimization of gridshells

J. Melchiorre [a], A. Manuello Bertetto [a], G.C. Marano [b,a,*], S. Adriaenssens [c]

[a] DISEG, Department of Structural, Geotechnical and Building Engineering, Politecnico di Torino, Corso Duca Degli Abruzzi, 24, Turin, 10129, Italy
[b] School of Civil Engineering and Architecture, Henan University of Technology, 100 Lianhua St, Zhongyuan District, Zhengzhou, Henan, 450001, China
[c] Department of Civil and Environmental Engineering, Princeton University, E208 Engineering Quadrangle, Princeton, New Jersey 08544, USA

ARTICLE INFO

Keywords:
Interactive optimization
Structural optimization
Artificial intelligence
Human-AI interaction
Human-in-the-loop

ABSTRACT

Recent advances in AI have improved automation and optimization capabilities. However, current systems remain limited in reproducing the intuitive and subjective aspects of human creativity. In architectural and structural design, this creates a gap between performance-driven optimization methods and the designer's qualitative intent, as existing workflows rarely integrate human preferences within a rigorous optimization process. This paper introduces a Human-in-the-Loop optimization framework that incorporates designer preferences as part of a multi-objective formulation. The method combines parametric modeling, evolutionary optimization and machine learning to guide the search toward solutions that balance efficiency, sustainability and creative intent. The framework is validated through a user study in which participants generate distinct Pareto fronts aligned with individual preferences. Results demonstrate that the approach effectively embeds human feedback within structural optimization, fostering more creative and informed exploration in preliminary design.

1. Introduction

In recent years, the research in artificial intelligence (AI) has increasingly focused on developing systems that are autonomous, adaptive and effective across different contexts. This trend is driven by the aim of achieving Artificial General Intelligence (AGI), which is intended to replicate human cognitive flexibility and reasoning abilities. In this context, computational creativity has emerged as a promising approach to AGI (Ventura et al., 2014). Modern AI is highly effective in structured tasks such as logical reasoning (Adeli & Park, 1996), problem solving (Entezami & Sarmadi, 2025; Li et al., 2025) and quantitative optimization (Park & Adeli, 1997). However, it remains limited in terms of creativity, a quality typically associated with humans, as it relies on intuition, cultural knowledge and subjective experience. Although AGI aims to be adaptable, able to solve problems across domains and make independent judgments (Pennachin et al., 2007), these capabilities alone do not represent creativity. Genuine creative capability requires domain knowledge and the ability to transcend it in novel and unexpected ways (Grba, 2020; Ventura et al., 2014). Extensive debate has been motivated by the question of whether creativity is a unique human trait or if it can also emerge in AI systems (Boden, 2004). Intelligence and creativity are widely recognized as being interconnected, although their relationship remains conceptually complex. Intelligence is typically defined as the capability to learn, apply knowledge and adapt to different situations (Legg & Hutter, 2007), while creativity can be considered as both an individual trait and an intrinsic property of processes, or as a skill that can be developed with practice (Simon, 1985).

Although the progress of AGI research may not necessarily lead to the development of creative machines, an increasing amount of research is investigating this possibility (Ventura et al., 2014). However, the idea of extending AI beyond data processing and reasoning toward human-like creativity and intuition raises important questions. Human creativity is deeply rooted in emotion, intentionality and cultural resonance, which are dimensions that current AI systems are not able to internalize. While large-scale neural networks can simulate or combine existing patterns (Adeli, 2001), they struggle to produce the fundamental conceptual innovation that characterizes human creativity (Zwir et al., 2022). This raises questions about whether it is appropriate to delegate decision-making tasks that require aesthetic judgement or cultural knowledge to data-driven systems. Rather than attempting to replicate human-like cognition, a more promising approach is to design collaborative frameworks that combine the computational precision of AI with human intuition and expertise. The objective is not to substitute human

* Corresponding author at: DISEG, Department of Structural, Geotechnical and Building Engineering, Politecnico di Torino, Corso Duca Degli Abruzzi, 24, Turin, 10129, Italy.
Email address: giuseppe.marano@polito.it (G.C. Marano).

https://doi.org/10.1016/j.cacaie.2026.100044
Received 10 October 2025; Accepted 7 March 2026
Available online 29 April 2026
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creativity, but rather to enhance it by extending the boundaries of design exploration.

This perspective has particular relevance in the fields of architecture and structural design, in which creativity is central to innovation. The design process is both a cultural and an aesthetic exploration, with roots in historical traditions and contextual adaptation. However, the majority of current optimization workflows consider architecture only as a fixed prerequisite, reducing it to a constraint rather than a generative driver (Thurston & Sun, 1994). In this decoupled approach, there is often a conflict between the architectural intent defined by the designer and the structural optimization process, resulting in suboptimal compromises. In contrast, an integrated framework allows creative intent and performance criteria to evolve together, resulting in a more adaptive process. The role of the designer is not marginal but guiding, interacting with computational tools to incorporate both quantitative measures and subjective qualities into the optimization process.

The objective of this paper is to introduce a Human-in-the-Loop (HitL) methodology that integrates designer preferences into structural optimization, by combining interactive optimization with machine learning (Adeli, 1995). In this context, the term Human-in-the-Loop indicates computational workflows in which human feedback is actively incorporated into algorithmic decision-making. HitL methods have been successfully applied in fields such as wearable robotics and assistive technologies, where user input is critical for tailoring solutions to individual needs (Díaz et al., 2023; Zhang et al., 2017). In structural design, early studies have already highlighted the creative potential of interactive optimization (Contiguglia et al., 2026). One of the first notable contributions was by Von Buelow (2008), who showed how evolutionary strategies can foster creativity by generating diverse sets of viable solutions rather than converging on a single optimum, encouraging broader exploration of the design space. Building on this foundation, Mueller and Ochsendorf (2015) and later (Danhaive & Mueller, 2015) developed frameworks that integrate designer preferences into structural optimization. Using interactive evolutionary algorithms, they allowed qualitative input to directly guide performance-based workflows, with designers influencing the search by selecting preferred solutions. More recently, the field has advanced with the adoption of machine learning and dimensionality reduction. Ochoa et al. (2021) combined self-organizing maps with gradient-boosted trees to establish a real-time feedback loop for solution selection. Their framework supports a post-typological approach, enabling exploration of structural forms beyond predefined categories.

Building on these foundational contributions, our work extends the paradigm of interactive structural optimization by unifying design space exploration, preference learning, and multi-objective optimization within a single Human-in-the-Loop framework. The proposed methodology provides a direct interaction between artificial intelligence and the designer, guiding the optimization through human intuition and creative intent (Melchiorre et al., 2024). This interaction is embedded in a multi-objective evolutionary framework in which one objective reflects architectural and functional preferences defined by the designer. Quantitative metrics are combined with qualitative input, while machine learning progressively learns from the designer to align the search strategy with their creative vision. The process starts from a parametric model controlled by the designer and connected to an interactive optimization algorithm that explores the design space, generating alternatives that balance performance and creative intent. The outcome is a Pareto-optimal set of solutions, enabling informed decisions and the selection of appropriate compromises in the conceptual design phase. The proposed methodology is not intended as a rigid framework of algorithms, but to promote a design philosophy that integrates computational efficiency with human creativity. The Human-in-the-Loop framework is not designed to replace traditional multi-objective optimization in terms of convergence or objective values. Instead, it structures the exploration and interpretation of Pareto-optimal solutions by explicitly incorporating designer intent while preserving quantitative optimality and diversity. In this work, the method is implemented in a form finding (Kociecki & Adeli, 2015; Manuello Bertetto et al., 2024) and optimization process for gridshell structures (Melchiorre et al., 2025, 2024) to demonstrate its ability to generate different solutions reflecting various designer preferences. Structural geometries are generated using a parametric approach combined with the Force Density Method (FDM) (Schek, 1974). Each generated structure is converted into images and processed by a Vision Transformer (ViT) for automatic feature extraction. To mitigate the curse of dimensionality (Köppen, 2000), Principal Component Analysis (PCA) (Greenacre et al., 2022) projects the features into a compact embedding space, where distances reflect visual and structural similarity. A fuzzy C-means clustering algorithm (Bezdek et al., 1984) is then applied to group similar solutions and a limited number of representative configurations from each cluster are evaluated by designers. Then, their feedback is generalized to the full set of solutions using the clustering structure. Alongside user preferences, economic cost and environmental impact are computed as additional objective functions. Structural feasibility is assessed through finite element analyses, using simplified verification criteria on displacements, stresses and local buckling. Finally, multi-objective optimization is performed using NSGA-III (Jain & Deb, 2014), which produces Pareto-optimal sets of solutions. To test the capability of the system to identify individual preferences, the procedure is repeated with feedback from 20 different decision makers, each generating a distinct Pareto front aligned with their individual preferences.

2. Interactive optimization

The general formulation of an optimization problem can be expressed as:

$$\begin{aligned} &\text{Find } \vec{x} = [x_1, \dots, x_n]^T \in \Omega \subseteq \mathbb{R}^n \text{ such that} \\ &\quad \min_{\vec{x}} \vec{f}(\vec{x}), \quad \text{where } \vec{f}(\vec{x}) = [f_1(\vec{x}), \dots, f_m(\vec{x})]^T \in \mathbb{R}^m \\ &\text{s.t.} \quad g_q(\vec{x}) \le 0 \qquad \forall q = 1, \dots, n_q, \\ &\quad h_r(\vec{x}) = 0 \qquad \forall r = 1, \dots, n_r, \end{aligned} \tag{1}$$

In this formulation, $\vec{x}$ defines the n design variables, bounded by box constraints $x_i^l \le x_i \le x_i^u$ that define the design space Ω. The objective function vector $\vec{f}(\vec{x})$ consists of m scalar functions $f_i(\vec{x})$, each representing a performance criterion to be minimized. Feasible solutions must also satisfy n_q inequality constraints $g_q(\vec{x}) \le 0$ and n_r equality constraints $h_r(\vec{x}) = 0$, which encode the structural and functional requirements of the problem. In general, interactive optimization problems preserve the classical multi-objective formulation but extend it by including an objective function defined according to the preferences of the decision maker.

The following sections introduce the main components of the proposed optimization problem. First, the parametric generation of structural configurations is described, outlining how different modeling parameters enable the production of architecturally heterogeneous solutions and provide the decision maker with a broad spectrum of alternatives. Next, the constraints are defined and grouped into two categories: geometric constraints and structural constraints. The role of the design variables is then discussed, highlighting how they control geometry, mechanics and material properties within the parametric framework. Finally, the objective functions are introduced, combining quantitative criteria with qualitative design preferences. These preferences are generalized through machine learning techniques that identify evaluation patterns, reducing the number of explicit user ratings while maintaining consistency with the intent of the designer.

2.1. Parametric form finding

This section describes the procedure adopted to generate the gridshell structures used in the optimization problem. A parametric design process, combined with a form finding procedure, enables the creation of a wide range of structurally heterogeneous configurations.

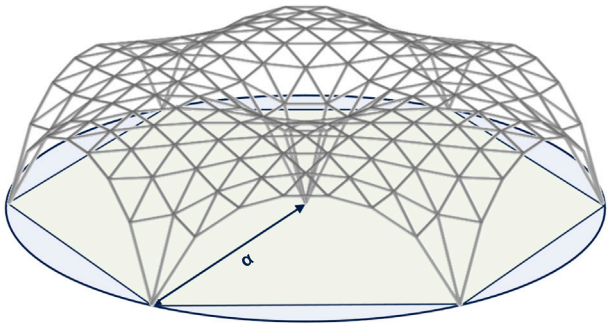


Fig. 1. Definition of the gridshell design problem, illustrating the circular area to be spanned, the inscribed base polygon used for generation and an example of the resulting gridshell structure.

The design problem involves creating a structure that spans a circular area with a radius of α, which in this study is set to 10 meters. The gridshell geometry is generated from a base regular polygon inscribed within a circular area (Fig. 1). The first design variable is the number of polygon sides n_{side}, while a second parameter controls the edge length of triangular elements, which defines mesh density and resolution. The internal surface is discretized through an automatic triangular meshing procedure using the pygmsh library in Python, enabling the generation of a wide variety of initial meshes (Fig. 2).

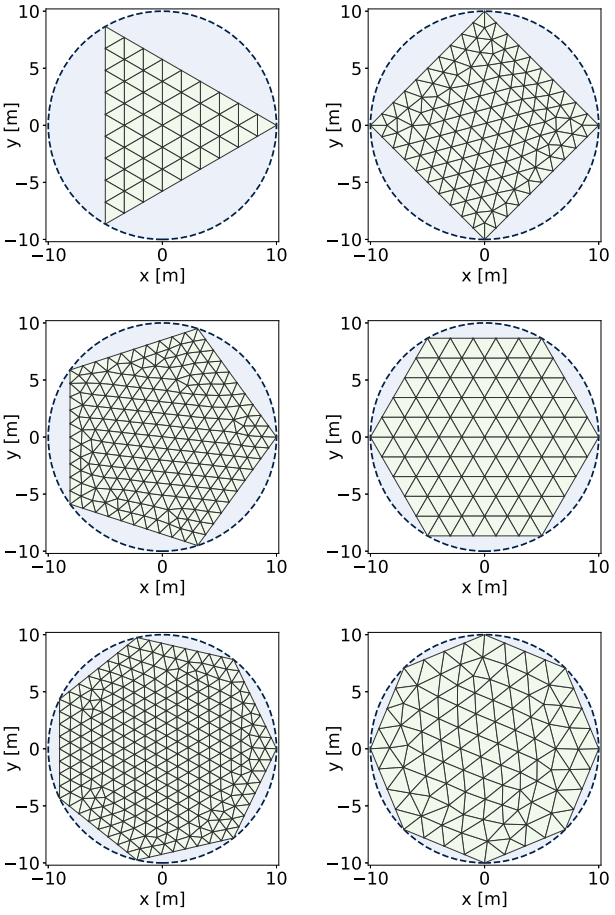


Fig. 2. Illustrative examples of meshed regular polygons inscribed in the circular domain. The mesh (dark gray) and polygon area (light green) are shown within the reference circle (blue). Variations are obtained by changing the number of polygon sides n_{side} and the mesh density.

Starting from this parametric mesh, the structural shape is obtained using the Force Density Method (FDM) (Linkwitz & Schek, 1971; Schek, 1974), a computationally efficient and widely used form finding technique for lightweight structures. The Force Density Method (FDM) assigns to each structural member a force density, defined as the ratio of internal force to element length. When the force densities are prescribed, the equilibrium equations can be reformulated as a linear system. Although these equations are typically nonlinear with respect to the nodal coordinates, this reformulation allows the direct computation of nodal positions, avoiding the need for iterative nonlinear analysis. FDM does not require material stiffness or elastic properties, relying only on geometry and prescribed force densities. This greatly reduces computational complexity, transforming the nonlinear equilibrium problem into a linear system of equations. Given its computational efficiency, the FDM is particularly suitable for Human-in-the-Loop protocols, where rapid generation of solutions is essential to maintain user engagement and avoid cognitive fatigue. By varying force densities q_j and boundary conditions a wide range of equilibrium configurations can be obtained. In this study, the Force Density Method is employed as a geometric form finding tool for early-stage design exploration. Consequently, stiffness-related effects, connection behaviour, serviceability checks and global stability phenomena are not explicitly addressed at this level and are considered outside the scope of the present example. These aspects are typically investigated in later design phases using higher-fidelity structural models. The proposed Human-in-the-Loop optimization framework is not linked to FDM and can easily incorporate alternative stiffness-aware or nonlinear form finding approaches, when a more detailed structural assessment is required.

2.2. Constraints

Structural optimization requires the definition of constraints that guarantee the feasibility and reliability of the generated solutions. In this study, these constraints are grouped into two categories: geometric and structural. In this work, the constraints are intentionally simplified in order to provide a generic and computationally efficient setting for constrained structural optimization. The proposed methodology is fully general and independent of the specific constraint definitions and can be extended to incorporate more detailed and application-specific constraints in real-world design scenarios.

The proposed optimization problem includes two geometric constraints to guarantee morphological and functional feasibility. The first limits the vertical extent of the gridshell, preventing excessively tall configurations. Specifically, the maximum node elevation z_{max} must not exceed the reference span radius of the covered area α:

$$z_{max} = \max_i(z_i) \leq \alpha = 10\,m \tag{2}$$

The second constraint requires that the perimeter edges of the structure remain sufficiently elevated above the ground to ensure functional use and accessibility. Let $\mathcal{E}_{\text{per}}$ denote the set of edges belonging to the structural perimeter. Each perimeter edge $e \in \mathcal{E}_{\text{per}}$ is defined by its two end nodes i and j, whose vertical coordinates are denoted as z_i and z_j, respectively. For each perimeter edge, the maximum nodal elevation is considered $z_{e,\max}$. The highest among all perimeter edges must not fall below a minimum clearance threshold $h_{\min}^{\text{edge}}$, which represents the minimum required vertical distance between the edge and the ground level:

$$z_{e,\max} = \max_{e\in\mathcal{E}_{\text{per}}} \left(\max(z_i, z_j)\right) \geq h_{\min}^{\text{edge}} = 2.5\,\text{m} \tag{3}$$

To express both constraints in standard form $g_q(\vec{x}) \leq 0$, they are rewritten as:

$$g_1 = \frac{z_{max}}{\alpha} - 1 \leq 0 \qquad g_2 = \tanh\left(\frac{h_{\min}^{edge}}{z_{e,\max}} - 1\right) \leq 0 \tag{4}$$

Table 1
Material properties used for structural simulations.

Material	E [GPa]	ν [-]	γ [kN/m^3]	f_y **[MPa]**
Structural Steel (S275)	210	0.30	78.5	275
Laminated Timber (GL24H)	12	0.35	6.2	24

In Eq. 4, the hyperbolic tangent function is introduced to obtain a smooth and bounded measure of constraint violation. This formulation preserves the feasibility condition while avoiding excessively large violations when $z_{e,\max}$ is small, which might otherwise dominate the constraint-handling mechanism in population-based optimization. The bounded and monotonic nature of the hyperbolic tangent ensures stable scaling of constraint violations without altering the feasible domain.

In addition to geometric limits, the optimization problem also includes structural constraints. A simplified verification framework is adopted to assess the feasibility of each candidate configuration during the conceptual design phase, avoiding reliance on computationally expensive simulations. These checks provide a compromise between computational efficiency and reliability in structural evaluation. The optimization process considers two materials: structural steel and laminated timber. Each material is defined by specific mechanical properties, which are summarized in Table 1.

In addition to material properties, each structural system adopts a different cross-section typology. For steel members, a non-standard Circular Hollow Section (CHS) is used, where the external radius r is a design variable and the wall thickness is set to:

$$t = 0.12r \tag{5}$$

For timber members, a solid rectangular section is assumed. The section height H is a design variable, while the base width b is defined as:

$$b = 0.5H \tag{6}$$

Following the assignment of material and section properties, a finite element model is developed for each candidate configuration. Structural analyses are performed using the OpenSeesPy (Zhu et al., 2018) framework, a Python interface for the OpenSees software.

Three structural requirements are then verified for each configuration.

$$\begin{aligned} &\delta_{max} < \delta_{lim} \\ &|\sigma_{max}| < f_y \\ &N_{max}^{Compression} < N_{Buckling} \end{aligned} \tag{7}$$

In this context, δ_{max} denotes the maximum vertical displacement, which must remain below the allowable limit δ_{lim}, set to $L/250$, where L is the largest free span of the structure. The second condition requires that the maximum stress σ_{max} does not exceed the material yield strength f_y. Finally, the axial compression force $N_{max}^{\text{Compression}}$ is checked against the Euler buckling capacity $N_{Buckling}$ to prevent instability failure.

These verifications are expressed using standard optimization notation, where inequality constraints are written as $g_q(\vec{x}) \leq 0$:

$$\begin{aligned} g_3 &= \frac{\delta_{max}}{\delta_{lim}} - 1 \leq 0 \\ g_4 &= \frac{|\sigma_{max}|}{f_y} - 1 \leq 0 \\ g_5 &= \frac{N_{max}^{Compression}}{N_{Buckling}} - 1 \leq 0 \end{aligned} \tag{8}$$

In this study, the verification procedures are limited to linear analyses and simplified checks. This allows for an efficient exploration of the design space and supports the integration of Human-in-the-Loop feedback within the optimization process. It is well known that the structural behaviour of gridshells is strongly influenced by geometric nonlinearity (Bulenda & Knippers, 2001), connection stiffness and stability phenomena (Bazzucchi et al., 2017b) such as snap-through and post-buckling response (Bazzucchi et al., 2017a; Manuello et al., 2016). These aspects become critical in later design stages and are extensively discussed in the literature (Melchiorre et al., 2024). While such nonlinear analyses are not included in the present work, the proposed methodology is fully compatible with their integration. Advanced nonlinear checks could be incorporated as additional constraints without altering the overall workflow. Therefore, the presented example should be interpreted as a proof-of-concept demonstrating the interaction between optimization, machine learning and designer feedback, rather than as a complete structural validation for gridshells.

2.3. Design variables

In this study, structural configurations are defined by nine design variables $d_1, \ldots, d_9$, which control both the initial geometry and the structural properties of each candidate solution. Every structure is generated from a parametric mesh based on a regular polygon inscribed in a circular base area, representing the gridshell footprint. The first two variables specify the geometry of the base polygon mesh. The number of polygon sides is expressed as:

$$d_1 = n_{side} \in [3; 10], \tag{9}$$

while the density of the triangular mesh is controlled by the predefined edge length:

$$d_2 \in [1.0; 5.0]\ m \tag{10}$$

Once the base mesh is defined, the boundary conditions for the form finding process are introduced. The vertices of the base polygon are fully constrained in all three directions, while the third design variable controls the behaviour of the unconstrained edge points along the perimeter:

$$d_3 \in 0, 1 \tag{11}$$

with $d_3 = 1$ indicating that edge points can move only vertically, and $d_3 = 0$ allowing full spatial freedom.

The fourth design variable d_4 defines the constraint condition of the central node of the mesh:

$$d_4 \in \{0, 1\} \tag{12}$$

with $d_4 = 1$ fixing the node to the ground and $d_4 = 0$ corresponding to an unconstrained node during the form finding process.

Once the geometry and constraints are set, the final structural shape is obtained through the FDM. At this stage, three additional variables are introduced. The fifth variable specifies the force density assigned to the internal elements of the mesh:

$$d_5 = q_{interneal} \in [0.1; 40.0]\ kN/m \tag{13}$$

The sixth design variable d_6 defines the ratio between the force densities of the edge and internal elements:

$$d_6 = \frac{q_{edge}}{q_{internal}} \in [0.5; 20.0] \tag{14}$$

This parameter controls the relative stiffness of the boundary elements, directly affecting the overall curvature and global shape of the gridshell.

Finally, the seventh design variable d_7 defines a vertical nodal loads applied uniformly to the mesh nodes, expressed as vertical force vectors $F = [0, 0, d_7]$:

$$d_7 = F_z \in [1.0; 40.0]\ kN \tag{15}$$

Variables d_5, d_6, and d_7 define the mechanical parameters of the form finding process, influencing the resulting geometry. Once the shape is

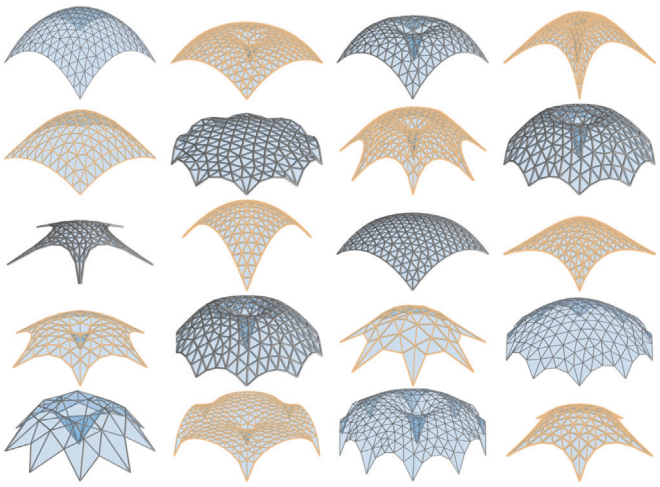


Fig. 3. Representative examples of gridshell structures generated by varying the design variables $d_1, \ldots, d_9$.

determined, two further variables describe the structural properties. The eighth design variable $d_8 = 0$ specifies the material:

$$d_8 \in \{0, 1\} \tag{16}$$

with $d_8 = 0$ for structural steel (S275) and $d_8 = 1$ for laminated timber (GL24h). The material choice also fixes the section typology: circular hollow sections (CHS) for steel and solid rectangular sections for timber. Finally, d_9 defines the cross-sectional size.

$$d_9 \in [0.01; 0.30]\ m \tag{17}$$

For steel members, d_9 defines the external diameter of the circular hollow section, while for timber it corresponds to the height of the rectangular section. Fig. 3 illustrates examples of configurations generated by varying the nine design variables. The combination of geometric, mechanical and material parameters enables the exploration of a wide solution space, offering the designer multiple alternatives for evaluation.

2.4. Objective functions

In this work, the optimization problem is formulated with three objective functions: economic cost, environmental impact and designer preferences. The first two provide quantitative measures derived from material quantities and properties, while the third captures qualitative evaluations to align solutions with architectural creativity and functional intent. Depending on project requirements, alternative or additional objectives could be considered, such as construction time, structural robustness or material reusability. In this study, economic and environmental metrics were selected because they are widely recognized and highly relevant performance indicators in structural design. The cost functions are computed by combining the FDM-generated geometries with material properties. The procedure evaluates the total material quantity and multiplies it by the corresponding cost coefficients for economic cost and environmental impact.

$$Costs = \sum_{material} SQL \times K \tag{18}$$

where SQL represents the material quantity and K the unit cost coefficient. For the economic cost, K is expressed in €/kg for steel and timber and in €/m^2 for glass panels. For the environmental impact, K corresponds to the embodied carbon, expressed in kgCO_2/kg.

The analysis considers three materials: structural steel (S275), glued laminated timber (GL24h) and double-glazed glass. Their cost-related parameters are listed in Table 2. Economic data are derived from the 2022 DEI Italian construction price list (Genio Civile, 2022), while environmental impact values are sourced from the Inventory of Carbon and Energy (ICE) database (De Wolf et al., 2016) and from the Carbon Leadership Forum (CLF) Baseline dataset of the EC3 platform (Waldman et al., 2023). Economic and environmental costs are expected to be strongly correlated in this formulation, as both objectives are computed from the same underlying material quantities using mass-based unit coefficients (Table 2). As a consequence, configurations with similar material distributions may exhibit nearly proportional trends between economic cost and embodied carbon. However, they differ since the use of either steel or timber introduces trade-offs in cost–performance relationships. In addition to these quantitative criteria, a third objective represents designer preferences. Incorporating designer input directly into the optimization loop allows the algorithm to evolve toward solutions that align not only with measurable performance criteria but also with the creative intent and qualitative evaluation. This overcomes a common limitation of automated workflows, which often produce technically valid solutions but fail to account for spatial, aesthetic or qualitative aspects that are difficult to quantify.

Table 2
Economic unit costs and embodied CO_2 factors for structural steel, laminated timber and double-glazed glass. Economic cost data are derived from the DEI Italian construction price list (Genio Civile, 2022), while embodied carbon values are taken from the ICE database and the EC3 Carbon Leadership Forum baseline dataset (De Wolf et al., 2016; Waldman et al., 2023).

Material	Cost [€/kg or €/m^2]	Embodied CO_2 [kgCO_2/kg]
Structural Steel (S275)	5.24 €/kg	1.27
Laminated Timber (GL24H)	4.46 €/kg	0.70
Double-glazed Glass	226.22 €/m^2	1.63

Qualitative feedback is incorporated into the optimization loop by considering the designer as a surrogate objective function. Each candidate solution is shown as an image and rated through a graphical user interface (GUI, Fig. 4), producing scalar scores that can be processed like any other objective function. The interface presents a three-dimensional visualization with interactive rotation and zooming. Structural materials are distinguished by colors providing a realistic and interpretable representation of each configuration. This enables the designer to assess both the overall geometry and the proportions of individual elements. For every proposed configuration, the designer provides a rating on a scale from 1 to 5. The optimization algorithm incorporates this score

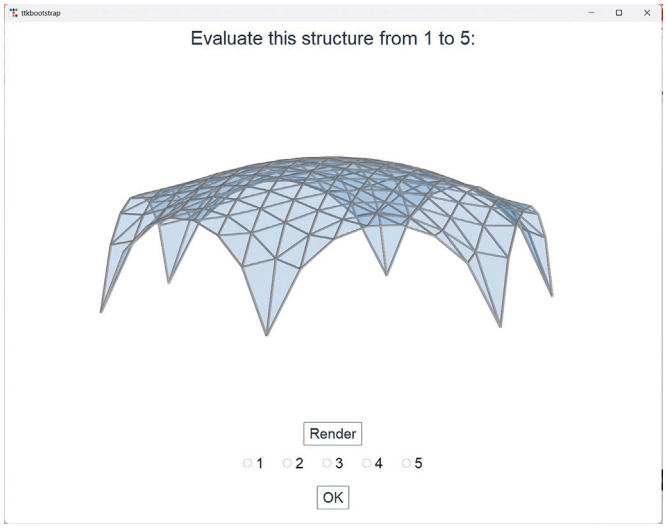


Fig. 4. Graphical user interface (GUI) for evaluation of the structural configuration during the optimization loop.

as a third objective, maximizing designer preference while minimizing economic cost and environmental impact. In this way, the designer provides the connection between qualitative assessment and computational evaluation, with numerical ratings translating subjective and often unmeasurable criteria into data that can be processed by the algorithm. As a result, the optimization process is informed not only by measurable indicators but also by human insight and expertise.

To support contextual evaluation, the framework integrates an AI-based rendering feature. Through the GUI, the designer can activate the "Render" function and provide a brief textual description of the structure and its surroundings. This input, combined with the geometric model, is processed through a deep learning pipeline to generate photorealistic, context-aware visualizations. The rendering system is based on state-of-the-art text-to-image diffusion models (Rombach et al., 2022), such as Stable Diffusion (Esser et al., 2024). These models combine three main components: a Variational Autoencoder (VAE) that maps images between pixel and latent spaces, a U-Net denoising network that reconstructs clean latent representations from noisy inputs, and a pretrained text encoder that aligns the generation with user-provided descriptions. During training and inference, Gaussian noise is added to the latent space and progressively removed by the U-Net, while cross-attention layers inject semantic information from the textual prompt (Vaswani et al., 2017). The VAE decoder then reconstructs the final image in pixel space, producing a realistic visualization of the structural configuration, aligning the generated image with the prompt description. Stable Diffusion provides semantic control through textual prompts, but it struggles to preserve precise spatial layouts which are a critical requirement in architectural and structural applications. To overcome this limitation, ControlNet extends diffusion models by introducing an additional branch that processes a reference image as a conditioning signal (Zhang et al., 2023). This branch, based on a trainable copy of the diffusion U-Net, predicts residual adjustments to the generative process, ensuring that the output respects both the textual description and the reference input. In this work, edge maps are used as conditioning images, as they capture the essential outlines of the structure while omitting unnecessary details. Edge maps are extracted with the Canny detector (Canny, 1986), which identifies continuous boundaries. Combined with textual prompts, these edge maps guide ControlNet to generate renderings that remain faithful to the structural geometry while embedding the design into the desired architectural context (Fig. 5). Secondary elements such as lighting effects, shadows and background details are not physically simulated and are generated heuristically by the diffusion model to provide a plausible visual context rather than a photometrically accurate rendering.

2.5. Search engine

In multi-objective optimization, the search engine is critical for exploring the design space and identifying the best solutions that represent trade-offs between conflicting objectives. Evolutionary algorithms, particularly the Non-dominated Sorting Genetic Algorithm (NSGA) (Deb et al., 2000), are commonly used to approximate the Pareto front from a set of candidate solutions. Conventional algorithms become less effective in problems with four or more objectives, commonly known as many-objective optimization. In such cases, the high-dimensional objective space reduces selection pressure and diversity, as most solutions are non-dominated. This makes it difficult to rank solutions effectively and to maintain a well-distributed approximation of the Pareto front. To address these challenges, NSGA-III was introduced (Deb & Jain, 2014; Jain & Deb, 2014). By introducing a set of reference directions, it improves convergence and ensures a more uniform distribution of solutions across the objective space.

In this study, NSGA-III was selected as the search engine for its ability to handle many-objective optimization problems. While the case study involves only three objective functions, the goal is to develop a general and scalable methodology. In this context, the NSGA-III provides the flexibility needed for future extensions. The implementation used here is the NSGA-III version available in pymoo (Blank & Deb, 2020), an open-source Python library for multi-objective optimization. NSGA-III is a metaheuristic optimization algorithm that preserves the evolutionary framework and non-dominated sorting of NSGA-II but replaces the crowding distance metric with a reference-point approach to maintain diversity. A predefined set of uniformly distributed reference directions is introduced in the objective space. During selection, candidate solutions are projected onto a normalized hyperplane and assigned to the closest reference direction based on perpendicular distance. This mechanism ensures a well-distributed set of solutions across the objective space. At convergence, each reference direction ideally corresponds to one Pareto-optimal solution, producing an even approximation of the Pareto front.

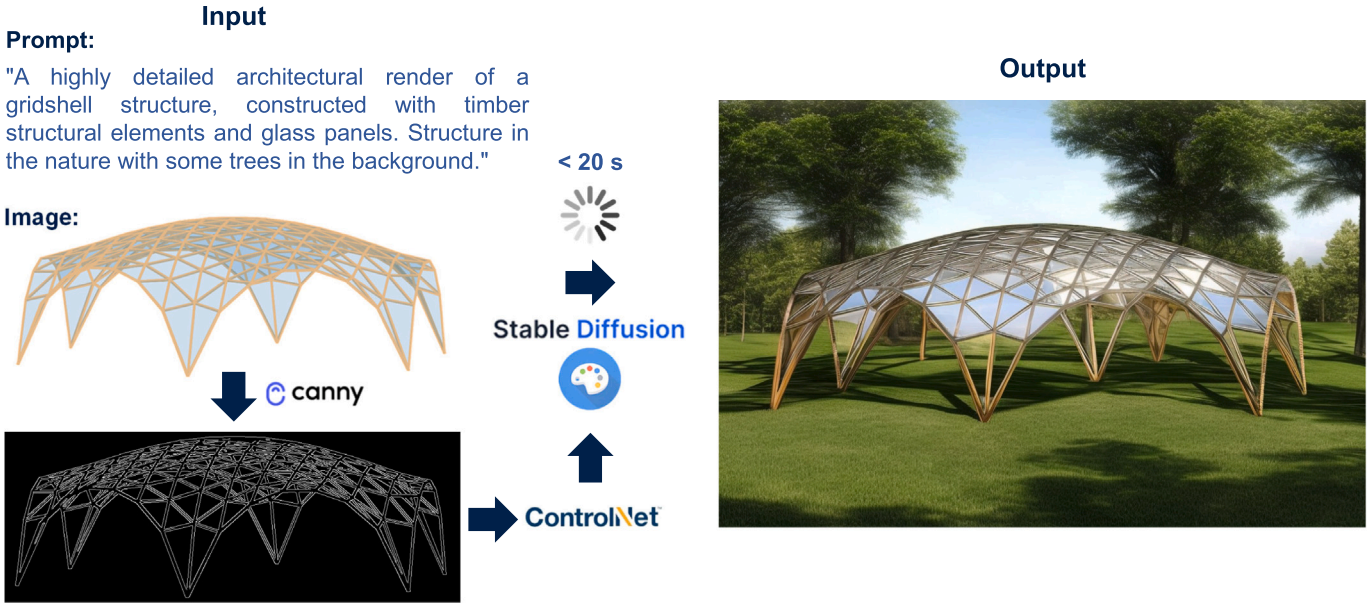


Fig. 5. Architectural rendering generation using stable diffusion with ControlNet and a Canny edge map.

2.6. Clustering

The interactive optimization problem, previously defined in terms of design variables, constraints, objectives and search engine, is addressed with a population-based evolutionary algorithm. This approach generates and evaluates many candidate solutions over successive iterations, gradually converging toward the Pareto front of optimal trade-offs. A critical limitation of such methods is their high computational cost which requires the generation and evaluation of a large number of candidate solutions. In interactive optimization, at least one objective relies on user feedback. When evolutionary algorithms generate thousands of candidate solutions, asking the decision maker (DM) to evaluate each one of them leads to cognitive overload, increasing the risk of inconsistent judgments caused by fatigue or shifting criteria over time. These inconsistencies introduce noise into the preference data, compromising the quality of the optimization and its ability to detect meaningful trends. To address this limitation, the methodology combines evolutionary optimization with machine learning. Specifically, a fuzzy clustering algorithm groups solutions according to visual similarity, so that only a few representative alternatives per cluster need to be rated by the DM. The remaining preferences are inferred from cluster memberships, reducing cognitive effort without losing representativeness.

In this work, clustering is performed on a fixed set of 100 candidate structural configurations generated through a Latin Hypercube Sampling (LHS) of the design space. This sampling strategy ensures a uniform coverage of the design variables and defines the initial population used both for feature extraction and for subsequent preference inference.

Clustering is performed based on visual similarity rather than predefined descriptive features, ensuring the general applicability of the framework across different structural typologies. A deep learning model is employed to automatically extract relevant features from images of the candidate solutions. Each structural solution produced by the optimization process is represented by three standardized images showing its principal views: top, front, and lateral (Fig. 6). These views are chosen to provide a complete visual characterization of each structure from different spatial perspectives. These images are processed with the pretrained ViT-B/16 model from PyTorch, which encodes the structural configuration into a high-dimensional embedding. The Vision Transformer is used exclusively as a pre-trained feature extractor. No task-specific training or fine-tuning is performed, resulting in minimal additional computational requirements for each design. Transformer-based architectures (Vaswani et al., 2017), and in particular Vision Transformers (ViTs) (Dosovitskiy et al., 2020), are powerful tools for image representation, capable of learning visual descriptors through self-attention mechanisms. Pretrained ViTs can produce dense embeddings that encode the essential content of an image, making them well suited for unsupervised tasks such as clustering. The ViT adapts the Transformer architecture, originally designed for language, to image data. The idea is to split an image into non-overlapping patches treated as tokens. In the ViT-B/16 variant, a 224×224 RGB image is split into 16×16 patches, producing 196 tokens. Each patch is projected into a D-dimensional embedding (with $D = 768$ in ViT-B/16) and enriched with positional information. Then, a special classification token [CLS] aggregates global information, providing a compact descriptor of the image.

In this work, each structural solution is represented by three orthographic views (top, front, lateral). Their embeddings, extracted with the ViT-B/16 model (768 dimensions per view), are concatenated into a single 2304-dimensional vector that encodes the structure (Fig. 6). While this representation is effective for capturing global morphological characteristics in early-stage design, it may be less accurate in representing fine-grained structural details or complex internal geometries that are not visually distinctive in the selected views. In any case, the distances in this embedding space reflect visual and structural similarity among candidate designs. After feature extraction, dimensionality reduction is applied to address the challenges of operating in high-dimensional spaces. Each candidate solution is represented by a 2304-dimensional embedding obtained from the three ViT-B/16 views. Direct clustering in such a space would be unreliable, as distance metrics lose discriminative power with increasing dimensionality. To address this issue, Principal Component Analysis (PCA) (Hotelling, 1933; Pearson, 1901) is applied. PCA transforms correlated variables into a smaller set of uncorrelated components that preserve most of the variance (Jolliffe & Cadima, 2016), improving computational efficiency and clustering reliability. In

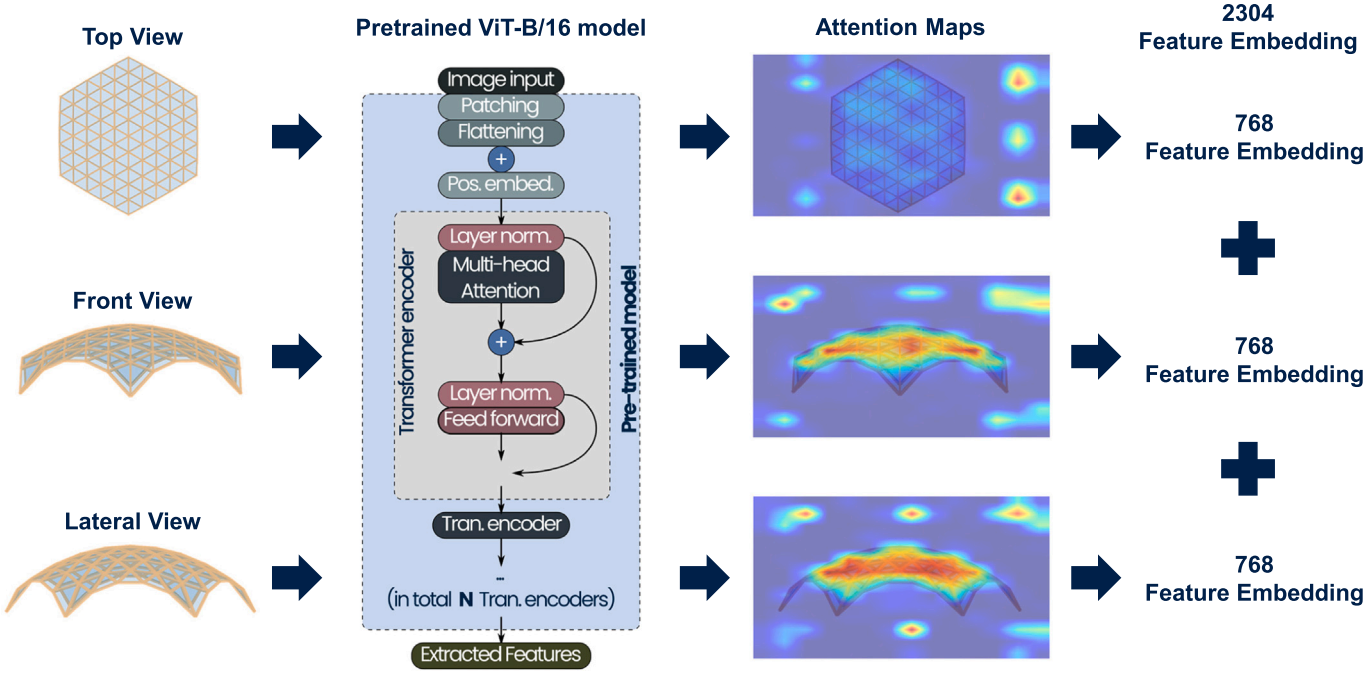


Fig. 6. Method illustration of the feature extraction process based on the pretrained ViT-B/16 model.

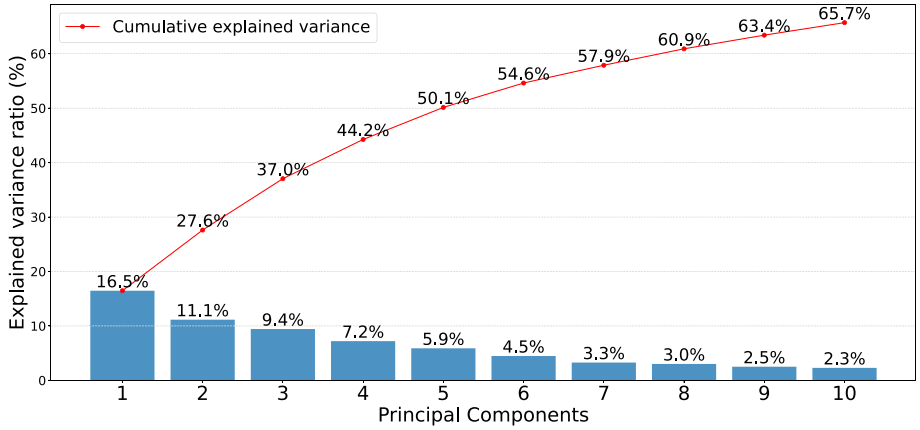


Fig. 7. Diagnostic plot for dimensionality reduction reporting the explained variance by the first 10 Principal Components, with cumulative variance shown in red.

this work, PCA was applied to a dataset of 100 candidate structural designs. These 100 designs define a fixed feature space on which PCA is computed once and subsequently reused for all clustering iterations, ensuring consistency across the inference process. The variance explained by the first ten components is shown in Fig. 7. For clustering, only the first five were retained, reducing dimensionality from 2304 to 5 ($\approx 99.8\%$ reduction) while still preserving over 50% of the total variance. Since each additional component contributed less than 5%, this choice provided an effective compromise between dimensionality reduction and preservation of relevant information.

The PCA-reduced feature vectors are then clustered using the Fuzzy C-Means (FCM) algorithm (Bezdek et al., 1984) to identify groups of visually related designs. This algorithm is particularly suitable for structural design problems where solutions often share characteristics across families. Unlike classical k-means, FCM assigns each design a degree of membership to multiple clusters, reflecting the fact that boundaries between structural typologies are not rigid. Formally, given N design embedding vectors $\vec{x} = \{\vec{x}_1, \dots, \vec{x}_N\}$, the algorithm divides the data into c clusters by computing a set of centroids $\vec{V} = \{\vec{v}_1, \dots, \vec{v}_c\}$ and a fuzzy membership matrix $U = [\mu_{ij}] \in [0,1]^{N \times c}$. Each μ_{ij} quantifies the degree of membership of design i in cluster j, subject to the normalization constraint:

$$\sum_{j=1}^{c} \mu_{ij} = 1 \quad \forall i \in \{1, \dots, N\} \tag{19}$$

The algorithm minimizes the following objective function, which generalizes k-means by weighting distances with fuzzy memberships:

$$J_m(U, V) = \sum_{i=1}^{N} \sum_{j=1}^{c} (\mu_{ij})^m \, \|\vec{x}_i - \vec{v}_j\|^2 \tag{20}$$

Here $|\vec{x}_i - \vec{v}_j|$ is the Euclidean distance, and $m \geq 1$ is the fuzzification coefficient controlling the softness of the clustering. For $m \to 1$ the method approaches hard clustering, while higher values allow smoother overlaps. In this study, $m = 2$ was adopted as a standard compromise. At convergence, the algorithm returns a set of cluster centers $\vec{V} = \{\vec{v}_1, \dots, \vec{v}_c\}$ and a membership matrix $U = [\mu_{ij}] \in [0,1]^{N \times c}$, where $\sum_{j=1}^{c} \mu_{ij} = 1$. Therefore, $\mu_{ij} = 1$ indicates full association with cluster j, while $\mu_{ij} = 0$ means no association.

To illustrate the process, 100 candidate configurations were grouped into $c = 10$ clusters. Table 3 presents representative configurations for the 10 clusters. Although FCM assigns non-zero memberships to multiple clusters, each design is shown with the cluster of highest membership. The examples correspond to the structures nearest to the centroids in PCA space, with the associated membership value reported below each image. This visualization supports the qualitative assessment of intra-cluster coherence and highlights the structural similarities identified by the clustering model.

2.7. Inference of the ratings

The fuzzy clustering of the candidate solutions forms the basis of the interactive optimization loop and of the Human-in-the-Loop inference process, designed to progressively estimate user preferences from a limited number of evaluations. The number of clusters c is not fixed *a priori* but is increased iteratively until prediction accuracy meets a predefined threshold. Prediction accuracy is assessed by comparing inferred ratings with the ratings provided by the designer for the subset of designs that have been evaluated at each iteration.

The procedure starts with an initial partition containing two clusters. At each iteration, a limited number of representative designs is selected from each cluster and submitted for the evaluation by the designer, while the remaining ratings are inferred. Increasing c refines the partitioning of the design space, improving discrimination between solutions at the cost of higher model complexity and evaluation effort. Specifically, the selected representatives include the design closest to the cluster centroid (high membership, $\mu_{ij} \approx 1$) and the most peripheral one (low membership, $\mu_{ij} \approx 0$). These evaluations capture both the core and variability of the cluster. Ratings are then propagated to all designs through a membership-weighted average.

For a cluster j containing a subset of designs $\mathcal{D}^j$, with user ratings $\mathcal{R}_i \in [1, 5]$ and memberships μ_{ij}, the cluster score S_j is computed as:

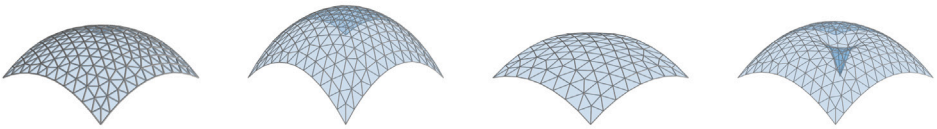
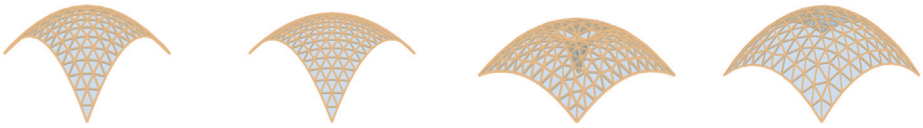
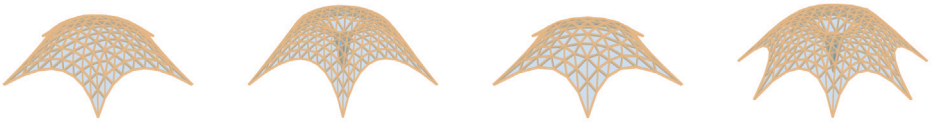
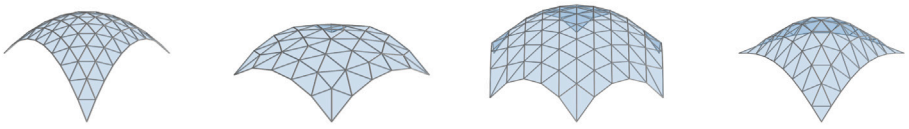
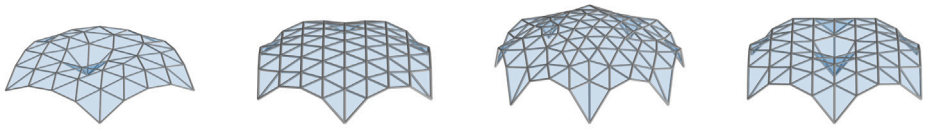
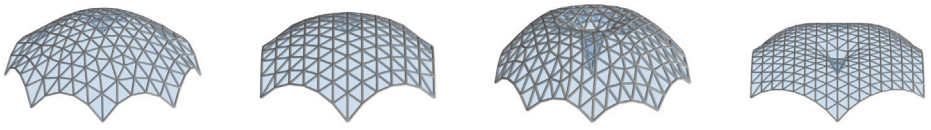
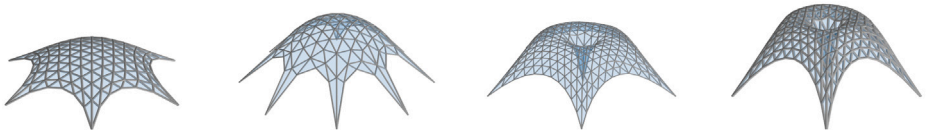
$$S_j = \frac{\sum_i \mu_{ij} \cdot \mathcal{R}_i}{\sum_i \mu_{ij}} \tag{21}$$

This formulation ensures that the cluster score represents the ratings of the most representative designs, while still accounting for the influence of peripheral members through their membership weights. The resulting cluster scores $S_1, \dots, S_c$ are then used to predict ratings for all designs. For each design i, the membership vector μ_i distributes its degree of association across the clusters, and the predicted rating $\hat{\mathcal{R}}_i$ is computed as:

$$\hat{\mathcal{R}}_i = \sum_{j=1}^{c} \mu_{ij} S_j \tag{22}$$

As a result, designs strongly associated with a single cluster inherit its score, while those with memberships distributed across multiple clusters receive an interpolated value reflecting their mixed affiliation. Through this fuzzy inference, subjective ratings can be estimated for the entire design set without requiring the decision maker to evaluate each individual solution. In practice, FCM serves as a surrogate model: structurally similar designs are grouped, a limited number of representative examples

Table 3
Illustrative clustering outcome showing representative structures from each of the 10 clusters, selected by maximum membership and visualized with their membership scores.

Cluster 1		$p = 82\%$	$p = 81\%$	$p = 62\%$	$p = 46\%$
Cluster 2		$p = 84\%$	$p = 83\%$	$p = 79\%$	$p = 71\%$
Cluster 3		$p = 87\%$	$p = 80\%$	$p = 79\%$	$p = 76\%$
Cluster 4		$p = 78\%$	$p = 74\%$	$p = 69\%$	$p = 67\%$
Cluster 5		$p = 72\%$	$p = 40\%$	$p = 31\%$	$p = 25\%$
Cluster 6		$p = 74\%$	$p = 61\%$	$p = 58\%$	$p = 56\%$
Cluster 7		$p = 97\%$	$p = 82\%$	$p = 80\%$	$p = 45\%$
Cluster 8		$p = 90\%$	$p = 82\%$	$p = 80\%$	$p = 76\%$
Cluster 9		$p = 90\%$	$p = 71\%$	$p = 53\%$	$p = 42\%$
Cluster 10		$p = 94\%$	$p = 44\%$	$p = 40\%$	$p = 40\%$

are rated by the user and these evaluations are propagated to the remaining designs using membership weights. As a result, user interaction is restricted to an initial inference phase involving a few evaluations. Preference scores are then automatically calculated for all remaining candidate solutions. The number of clusters is gradually increased until the predicted ratings $\hat{\mathcal{R}}_i$ align with actual designer feedback, ensuring that meaningful distinctions among solutions are captured while reducing evaluation effort.

A critical challenge is determining the appropriate number of clusters, as this directly influences both predictive accuracy and the cognitive workload of the user. Several methods have been proposed in the literature, including the elbow method, silhouette scores and the Fuzzy Partition Coefficient (FPC). These indices typically evaluate cluster compactness and separability, but in this work they proved inadequate for the specific requirements of the interactive framework. For this reason, a set of custom stopping criteria was introduced to ensure predictive accuracy while controlling model complexity, computational cost and the cognitive effort required from the designer.

1. **Empty Cluster Detection:** The presence of one or more empty clusters indicates that the subdivision is too fine-grained or unstable. In this case, further increasing the number of clusters does not improve structural differentiation and the iteration is terminated.
2. **Maximum Cluster Count:** An upper bound c_{max} is imposed on the number of clusters to control model complexity. In this study, c_{max} is set to 50.
3. **Maximum User Evaluations:** The number of ratings directly requested from the user is limited to $MaxUserEvaluations = 100$, in order to keep the cognitive burden at an acceptable level.
4. **Outlier Detection:** For each cluster j, the Euclidean distance between members and the cluster centroid is computed as:

$$d_{ij} = \|\vec{x}_i - \vec{v}_j\| \quad \forall\, i \;\; | \;\; \mu_{ij} > \mu_{ik},\; k \neq j. \tag{23}$$

To identify anomalous samples within each cluster, a robust, distance-based criterion is applied to the set of in-cluster distances $\{d_{ij}\}$. In particular, a sample is classified as an outlier if:

$$d_{ij} > Q_3 + \kappa\,(Q_3 - Q_1), \tag{24}$$

where Q_1 and Q_3 denote the 25th and 75th percentiles of the distance distribution, respectively, and $(Q_3 - Q_1)$ represents the interquartile range. The parameter κ is a factor that controls the strictness of the outlier detection and is set to $\kappa = 1.5$. Outliers with unusually large distances indicate fragmentation or poor cohesion, suggesting that further refinement is unnecessary.

5. **Predictive Accuracy:** The most critical criterion assesses the accuracy of the fuzzy inference model by comparing predicted ratings with newly provided user evaluations. The clustering process is terminated once a sufficient level of agreement is achieved. Specifically:
   - Mean Absolute Error (MAE) within each cluster:

$$\mathrm{MAE}_{\mathrm{cluster}} \leq 0.5 \tag{25}$$

   - Maximum Absolute Error (MaxAE) across all designs:

$$\mathrm{MaxAE}_{\mathrm{global}} \leq 1.0 \tag{26}$$

   If both thresholds are satisfied for five consecutive iterations, the clustering is considered stable.

3. Workflow

The overall workflow of the proposed methodology is illustrated in Fig. 8. It integrates parametric structure generation, feasibility checks, multi-objective optimization and Human-in-the-Loop preference inference into a unified process. Candidate gridshell configurations are first parametrically generated and evaluated with respect to geometric and structural constraints. Feasible solutions are then assessed with respect to both quantitative cost-related objectives and qualitative designer preferences. To minimize evaluation effort, the framework combines deep learning-based feature extraction with machine learning clustering methods, allowing designer ratings from a limited set of examples to be generalized across the entire solution space. Once the preference model is inferred, the optimization proceeds fully automatically, without further user interaction. These inferred scores are incorporated into the optimization process using NSGA-III, which evolves the population toward Pareto-optimal fronts balancing cost, performance and qualitative intent. Finally, the resulting Pareto set is visualized and presented to the designer, who makes the ultimate selection among the compromise solutions. This closes the optimization loop by integrating the designer into the process, ensuring that the final solutions are not only quantitatively optimal but also consistent with qualitative preferences. The individual computational components employed in this workflow, including deep feature extraction (Dosovitskiy et al., 2020),

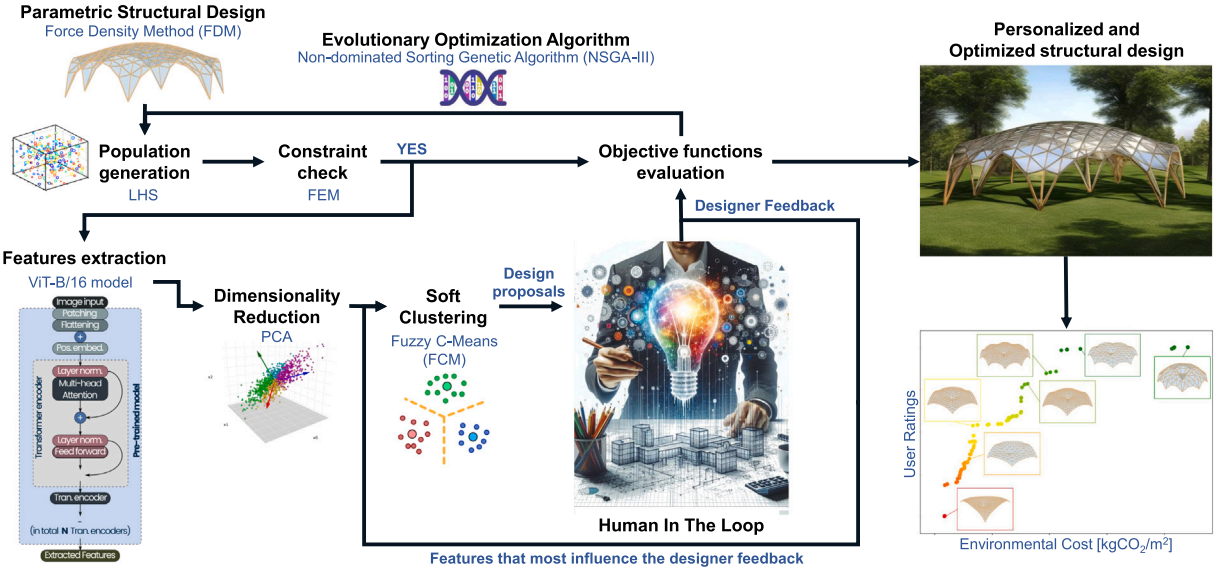


Fig. 8. Human-in-the-Loop structural optimization workflow.

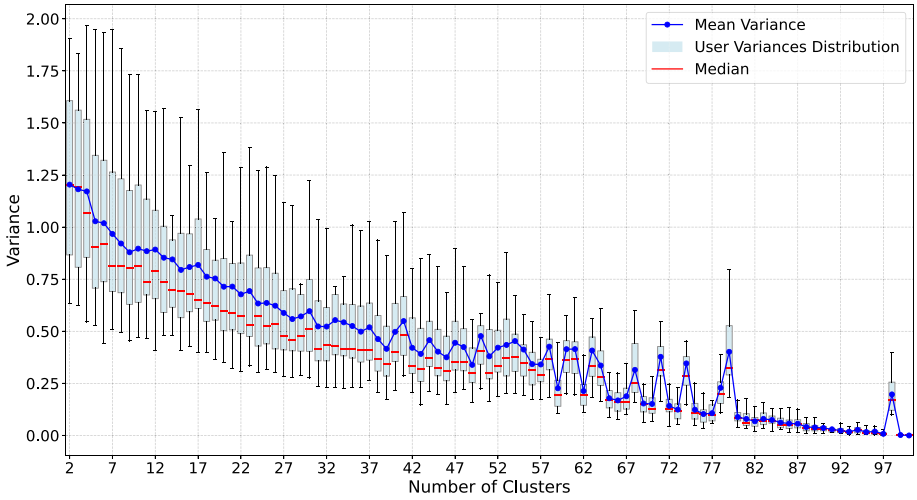


Fig. 9. Box plots of mean intra-cluster rating variance across users for different numbers of clusters, ranging from 2 to 100. Each box summarizes the distribution of average variances computed per user.

fuzzy clustering (Bezdek et al., 1984) and evolutionary multi-objective optimization (Deb & Jain, 2014), are well established approaches. The main contribution of this work lies in integrating these methods within a unified Human-in-the-Loop framework. This integration enables preference inference and a stable optimization based on subjective user feedback. The contribution is therefore methodological and focused on early-stage structural design.

4. Preference survey

The proposed methodology was validated through a survey involving 20 participants, each asked to rate 100 structural design alternatives on a five-point scale, from 1 (least preferred) to 5 (most preferred). The survey was conducted under strict anonymity. No personal identifiers were recorded and the information collected was entirely non-sensitive, limited to preferences on structural alternatives. Participants were not required to have expertise in structural design, although most participants had a technical background, primarily in civil engineering, architecture or related engineering disciplines. Data were collected in both in-person and remote sessions to better simulate realistic usage scenarios. In addition to collecting preference data, the survey responses were used to simulate the Human-in-the-Loop inference process described in Section 2.7. Although each participant rated all 100 structural designs, these complete evaluations were not directly used as input to the inference model. Instead, they constitute a reference dataset from which user evaluations are progressively retrieved to emulate realistic interaction with a designer-in-the-loop. At each iteration of the simulated inference process, only a limited subset of designs is treated as "evaluated" by the user, following the representative selection strategy described in Section 2.7. The corresponding ratings are extracted from the survey dataset and used to build the preference inference model, while the remaining ratings are considered unknown and inferred through clustering. This procedure allows the Human-in-the-Loop workflow to be tested in a controlled, repeatable and fully reproducible manner, without requiring repeated real-time user interaction.

The evaluation protocol was designed to promote consistency in user responses, since predictive accuracy depends on comparable designs receiving similar ratings. Although the methodology can tolerate a degree of inconsistency, its performance improves when evaluations are coherent. The survey included 100 structural solutions, a number chosen to balance solution diversity with required cognitive effort. This ensured representation of all structural typologies, with at least one cluster per type and a minimum of two ratings per cluster. A smaller dataset would have failed to capture sufficient variability, while a much larger one would have imposed excessive cognitive burden on the user. Although rating 100 solutions was still demanding and may have introduced some bias, the collected data provide a solid basis to evaluate the effectiveness of the proposed method. The first validation analysis examines whether increasing the number of clusters reduces the variance of user ratings within each cluster. Lower intra-cluster variance indicates that centroid scores are more reliable, improving the inferred ratings of unevaluated solutions. The dataset was clustered from 2 to 100 groups, and for each configuration the mean intra-cluster variance was computed across the 20 participants. Results are summarized in Fig. 9, where box plots illustrate the distribution of variances and how cluster count affects rating consistency.

Fig. 9 shows that local maxima in intra-cluster variance occasionally occur, indicating that for certain cluster numbers dissimilar solutions are grouped together. Nonetheless, the overall trend confirms the effectiveness of the clustering process, as intra-cluster variance steadily decreases with an increasing number of clusters. Importantly, the boxplot representation captures the distribution of intra-cluster variance across all participants, thereby explicitly accounting for inter-user variability in subjective ratings. The fact that median values are often below the mean further suggests that the observed fluctuations are primarily driven by a limited number of outlier users, rather than by systematic shortcomings of the clustering strategy.

The survey was also used to test the effectiveness of the proposed stopping criteria, designed to balance inference accuracy with the cognitive effort required from users. While increasing the number of clusters and collecting more evaluations reduces intra-cluster variance and improves prediction accuracy, it also increases evaluation effort. The stopping criteria mediate this trade-off, ensuring the process remains accurate yet efficient. To evaluate the stopping criteria, a simulated inference process was performed using the survey data. User ratings were treated as unknown and the number of clusters was progressively increased while inferring preferences at each stage. For each cluster, the ratings of the most central and most peripheral designs were retrieved from the survey dataset to simulate real user evaluations at that iteration. The process continued until a stopping condition was reached.

The predictive accuracy stopping criteria were defined in terms of a cluster-level threshold, $\mathrm{MAE}_{\mathrm{cluster}}$, and a global threshold, $\mathrm{MaxAE}_{\mathrm{global}}$, as introduced in Eq. 25 and Eq. 26. These criteria are intentionally conservative, in order to stress-test the inference process and to quantify the evaluation effort required to achieve reliable preference prediction. To further investigate the robustness of these stopping criteria, a sensitivity analysis was conducted by varying both the accuracy thresholds. Specifically, three values were considered for $\mathrm{MAE}_{\mathrm{cluster}}$ (0.3, 0.5 and

Table 4
Average number of user evaluations required to satisfy the predictive accuracy stopping criteria (mean ± standard deviation), as a function of the cluster-level threshold $MAE_{cluster}$ and the global threshold $MaxAE_{global}$.

$MaxAE_{global}$	$MAE_{cluster} = 0.3$	$MAE_{cluster} = 0.5$	$MAE_{cluster} = 0.8$
0.5	46.9 ± 18.8	41.0 ± 17.9	40.5 ± 17.7
1.0	47.5 ± 19.2	36.3 ± 15.9	26.5 ± 11.0
1.5	46.9 ± 18.8	35.9 ± 15.4	24.4 ± 9.0

0.8) and three values for $MaxAE_{global}$ (0.5, 1.0 and 1.5), resulting in nine different configurations. For each configuration, the inference procedure was simulated for all users, and the number of evaluations required to reach convergence was recorded. Table 4 reports the average number of evaluations together with the corresponding standard deviation across all simulations.

As expected, a higher level of accuracy requires more evaluations, while a less stringent threshold reduces the evaluation effort. Notably, even under the most conservative configurations, the average number of evaluations remains below 50, suggesting that the proposed inference strategy is robust. The thresholds adopted in this study represent a compromise between prediction accuracy and cognitive effort. For this configuration, the value of the mean plus one standard deviation is close to 50 evaluations, which was considered reasonable for user interaction. This metric accounts not only for average behaviour, but also for inter-user variability and stochastic effects in the inference process.

Fig. 10 shows the distribution of evaluations required for each user. For each participant, the inference process was simulated 300 times, each time starting from a different random initialization of the clustering algorithm. This results in a distribution of required evaluation counts per user, which is summarized using box plots to capture variability due to stochastic effects in the inference process. For 18 out of 20 participants, the median remained well below the target threshold of 50, and in all cases under 60. Variability was observed, with some users requiring fewer than 30 evaluations and others close to 60. The absolute numbers are influenced by the robustness condition, which enforces five consecutive accurate iterations before stopping. Since each iteration adds at least two new evaluations, this rule introduces a minimum of ten additional evaluations, with a stronger impact when the total number is limited. On average, the system reached the target accuracy with 36 evaluations per user. Aggregated results confirm this efficiency, considering that the global distribution has a median slightly above 30, and about 75% of simulations required fewer than 45 evaluations. These findings show that the methodology consistently achieves accurate preference inference while keeping cognitive effort at a manageable level. It should be noted that, when user evaluations are highly inconsistent, the stopping criteria naturally require a larger number of evaluations.

In addition to the number of required evaluations, the predictive accuracy of the inferred preference models was assessed using the Root Mean Squared Error (RMSE). For each user, the RMSE was computed for all 300 simulated inference runs by comparing inferred ratings with the corresponding survey ratings over the full set of designs. Fig. 11 reports the resulting box plot distributions, where each box summarizes

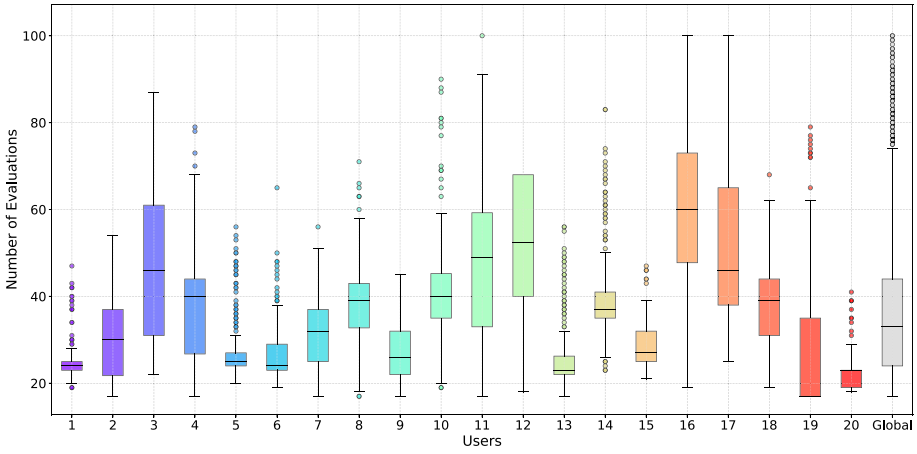


Fig. 10. Box plot distribution of required evaluations per user. For each of the 20 participants, the median, interquartile range and potential outliers are reported.

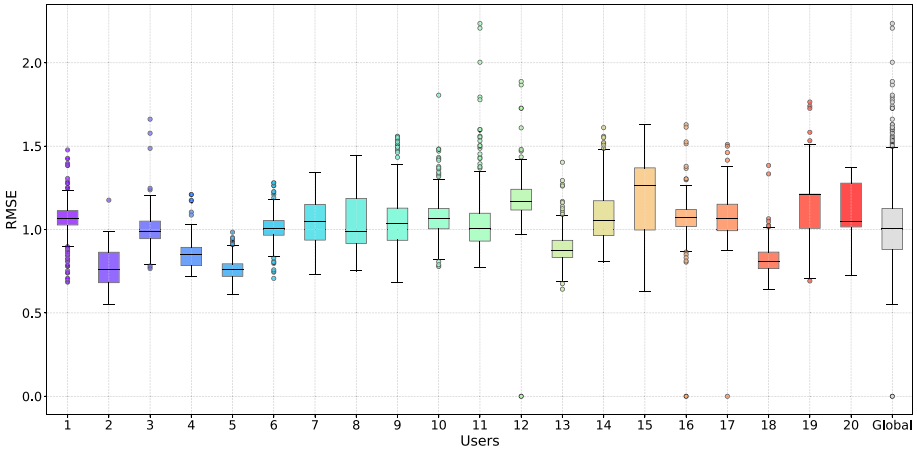


Fig. 11. Box plot distribution of RMSE per user. For each of the 20 participants, the median, interquartile range and potential outliers are reported.

Table 5
Evolutionary optimization settings used in the NSGA-III implementation.

Category	Parameter	Value
Initialization	Sampling method	LHS
	Population size	100
Evolution	Number of generations	50
Variation	Crossover operator	SBX ($p_c = 0.9, \eta_{SBC} = 20$)
	Mutation operator	Polynomial ($\eta_{mutation} = 30$)

the variability in inference accuracy due to stochastic clustering initialization. Across users, the median RMSE is approximately 1.0 on a 1–5 rating scale, indicating that the proposed method achieves a satisfactory level of accuracy despite relying on a limited number of explicit user evaluations. In the graph, the presence of occasional low-RMSE outliers corresponds to rare cases in which the inference process required a very large number of evaluations, leading to near-perfect agreement.

5. Optimization results

This section presents the outcomes of the optimization process based on the preference inference models derived from the user survey. A different model for each user was used to estimate designer preferences across the full set of candidate solutions. The inferred ratings are incorporated as a third objective function within the multi-objective optimization framework. The goal is to evaluate how user-specific preferences generate different sets of optimal solutions. The initial population was generated using Latin Hypercube Sampling (LHS) to ensure a uniform and unbiased coverage of the design space across all design variables. All optimization runs were performed using the NSGA-III algorithm with a fixed population size and number of generations. The main hyperparameters adopted for the evolutionary process, including crossover and mutation settings, are summarized in Table 5.

Fig. 12 shows Pareto fronts for four users, projected into two-dimensional subspaces for readability. The distinct patterns highlight how individual preference models influence the optimization process. As shown in Fig. 12, Pareto fronts differ significantly across users in both cost–preference and impact–preference spaces, confirming that inferred preferences directly influence optimization outcomes. Fig. 12(a) and (b) are very similar, indicating that economic and environmental costs are correlated. This relationship is examined in Fig. 13, which presents a scatter matrix of the three objectives for User 1. Off-diagonal scatter plots show pairwise relationships, while diagonal histograms illustrate the distribution of each objective. This visualization helps identify trade-offs when objectives are in conflict and synergies when improvements are aligned, offering insight into the structure of the Pareto front. In particular, the scatter plots of the two cost objectives reveal an almost linear relationship, indicating a strong positive correlation rather than a conflict. This can be explained by the fact that both economic and environmental costs are defined as weighted sums of material quantities, with weights given by unit prices or CO_2 emissions. In principle, the two functions should be independent, especially given the inclusion of steel and laminated timber with different parameters. However, in this case study, the structures span a limited radius of 10 meters, where the lower strength of laminated timber is still sufficient to withstand the applied loads. In this range, timber proves consistently more advantageous, being not only structurally adequate but also cheaper and more sustainable than steel. As a result, most of the optimal solutions adopt timber and the Pareto fronts collapse into curves rather than surfaces. Larger-scale structures could promote steel usage, but this would require much denser meshes and higher computational costs, which were incompatible with the parametric repeated analyses and optimization runs needed in this work. It is important to highlight that the examples used in this study are purely illustrative. The main focus of this study is to introduce and test a Human-in-the-Loop optimization framework, rather than

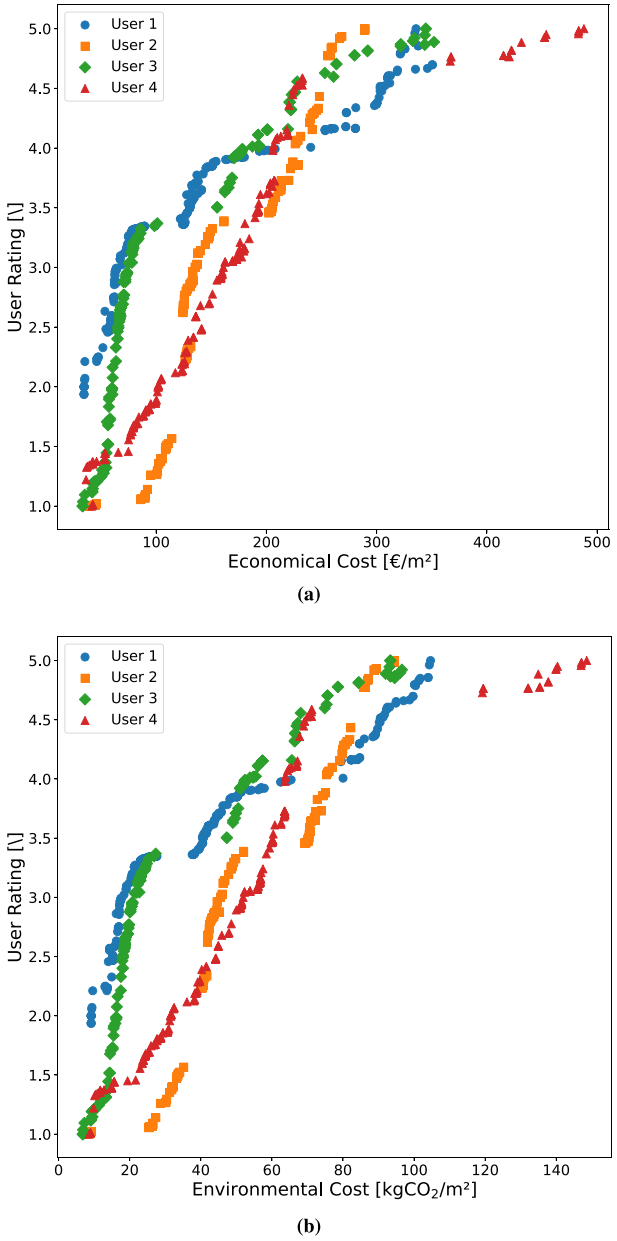


Fig. 12. Pareto fronts for users 1 to 4. Subfigure (a) shows the trade-off between economic cost [€/m^2] and inferred user ratings, while subfigure (b) presents the trade-off between environmental impact [kgCO_2/m^2] and user ratings. The different shapes of the Pareto fronts demonstrate the influence of individual preference models.

recommending a specific structural typology or metric. While the case study focused on a specific scale and set of objectives, the methodology remains general and can be extended to other systems and performance criteria.

Figs. 14–16 show the Pareto fronts for different participants, illustrating the trade-off between environmental impact (kgCO_2/m^2) and inferred user ratings. Each plot also visualizes a set of representative configurations from the corresponding Pareto front, enabling designers to interpret the results, explore trade-offs and identify promising solutions during the conceptual design phase. An important result is that designs with similar user ratings are often different in environmental impact. This demonstrates the value of the methodology, which integrates user preferences without restricting the design space a priori. Rather than limiting the parametric model to anticipated choices, the

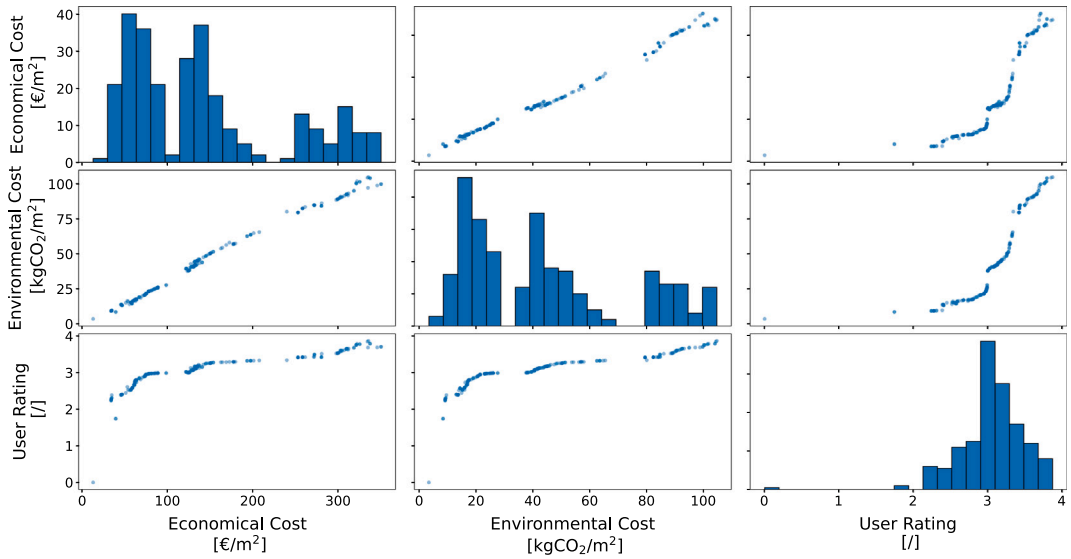


Fig. 13. Scatter matrix showing pairwise relationships among the three objectives: economic cost [€/m^2], environmental impact [kgCO_2/m^2] and user rating [–], inferred for user 1. Off-diagonal plots show bivariate relationships while diagonal plots show marginal distributions.

framework enables exploration of diverse typologies, revealing alternatives that may both align with user preferences and outperform others in quantitative objectives. Conversely, excluding user preferences from the optimization process could result in solution sets that are quantitatively optimal, but not aligned with the expectations of the designer. In contrast, overly restrictive parametric models based on user assumptions could prevent the discovery of better alternatives. The proposed approach shows how preference inference expands the design space and supports interactive evaluation of trade-offs. For example, in the case of User 20, a preference for steel led to steel-based designs being included in the Pareto front despite their higher environmental impact. At the same time, visually similar timber alternatives with lower impacts also appeared in the front, allowing direct comparison that would not have been possible with a more rigid framework. The results highlight that solutions with three supports often minimize material use, but many users favored alternatives perceived as more stable or aesthetically coherent. The framework makes these trade-offs explicit, generating solution sets that are both structurally and environmentally effective while reflecting individual preferences. The visual analysis of Pareto fronts confirms that user input directly shapes the optimal set, ensuring the decision-maker remains central to a context-aware exploration of structural design spaces.

To further evaluate the robustness of the proposed methodology in relation to the definition of economic and environmental objectives, an additional optimization scenario was considered, in which the unit cost coefficients of steel and laminated timber were modified to introduce a conflicting relationship between the first two objective functions. Specifically, the economic cost coefficient of steel was increased by one order of magnitude, while the cost coefficient of laminated timber was reduced by the same factor. In this scenario, steel became more favorable from an economic point of view while remaining more impactful from an environmental perspective. Fig. 17 shows the resulting Pareto front for User 1. The plot represents the trade-off between economic cost and environmental impact, respectively measured in [€/m^2] and [kgCO_2/m^2], while the inferred user rating is encoded through a color scale. The figure demonstrates that the qualitative influence of user preferences

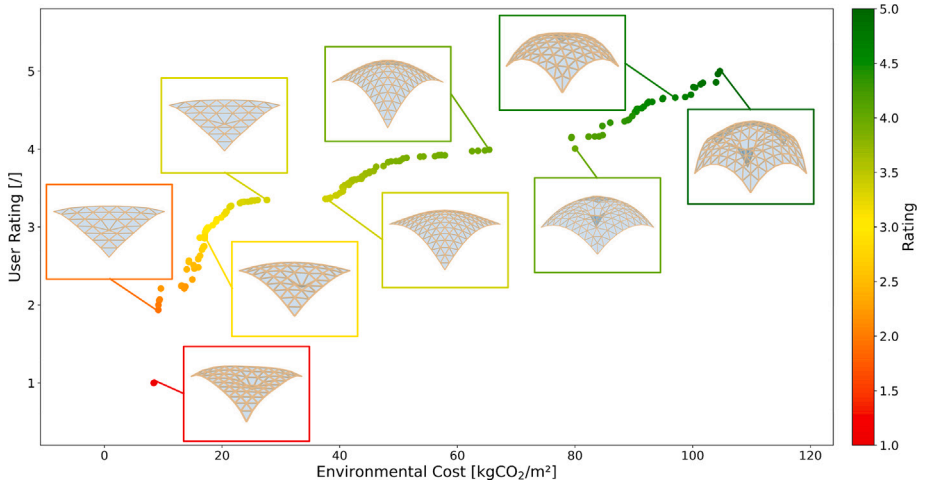


Fig. 14. Visual representation of the Pareto front showing the set of optimal structural solutions that illustrate the trade-off between environmental cost and the preference ratings of user 1.

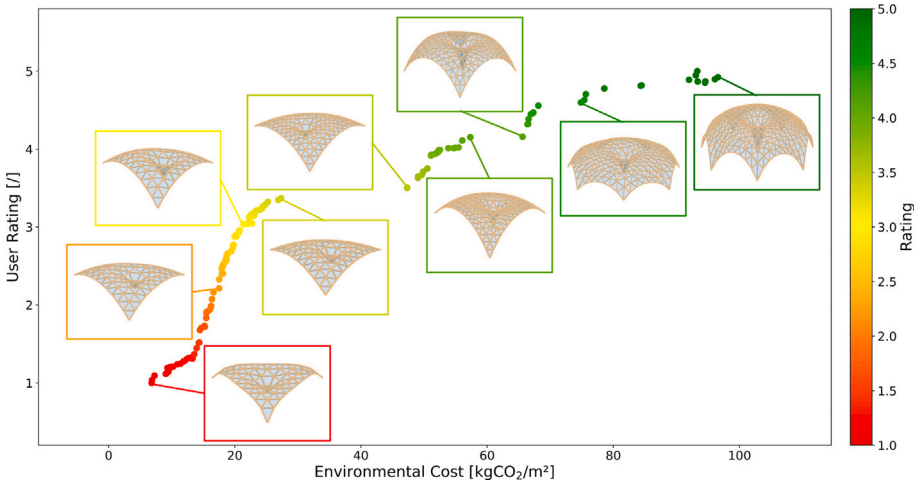


Fig. 15. Visual representation of the Pareto front showing the set of optimal structural solutions that illustrate the trade-off between environmental cost and the preference ratings of user 3.

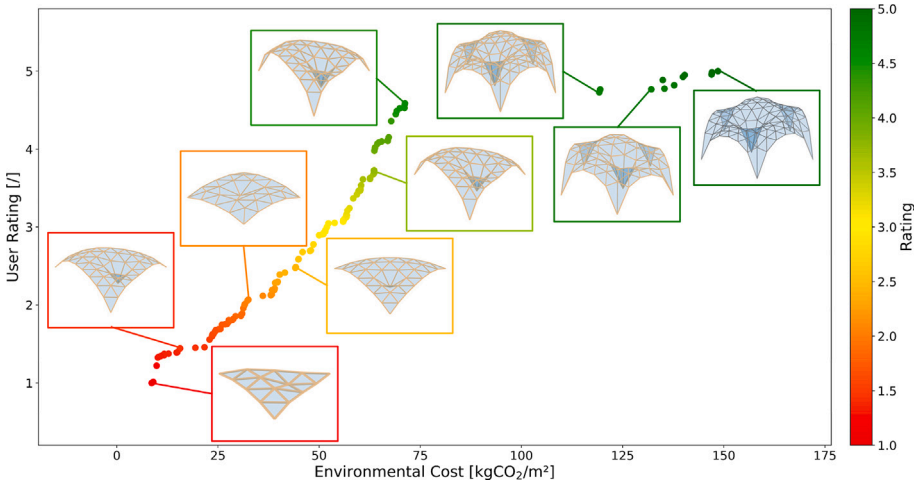


Fig. 16. Visual representation of the Pareto front showing the set of optimal structural solutions that illustrate the trade-off between environmental cost and the preference ratings of user 20.

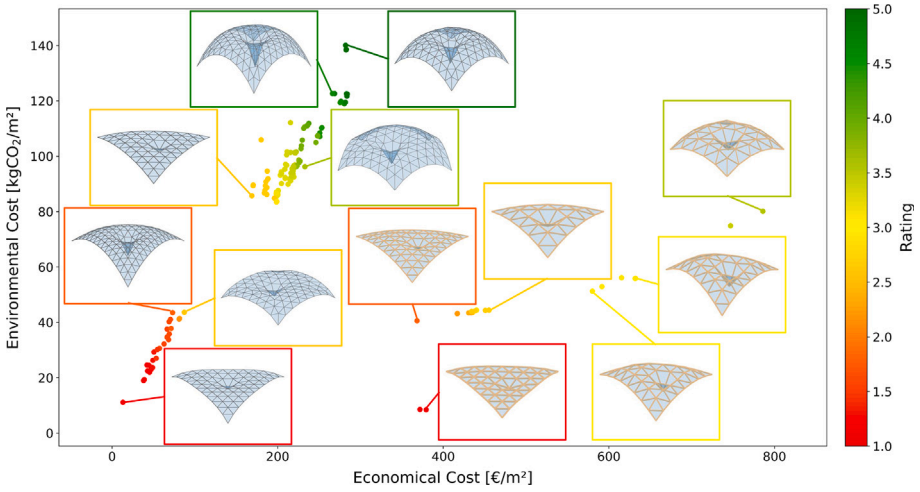


Fig. 17. Visual representation of the Pareto front for user 1 under the modified-cost scenario, showing the trade-off between economic cost and environmental impact. Inferred user ratings are represented through a color scale.

remains consistent. A comparison with the original optimization results shown in Fig. 14 reveals that solutions that achieve higher user ratings in the modified-cost scenario are characterized by morphologies that are qualitatively similar to those previously preferred, despite differences in material choice and cost assumptions. This qualitative consistency across different economic assumptions confirms that the optimization is effectively guided by the inferred user preferences, rather than the specific adopted cost or impact coefficients.

6. Conclusions

This work presents a Human-in-the-Loop framework for structural optimization that explicitly integrates designer preferences within a multi-objective evolutionary process. By embedding subjective feedback as an objective function alongside structural performance and environmental impact, the proposed methodology transforms optimization from a purely algorithm-driven task into a designer-guided exploration. The results obtained from a user survey demonstrate that different designers lead to distinct Pareto-optimal fronts, confirming the ability of the framework to capture individual preferences while preserving solution diversity and engineering reliability. Rather than proposing a fixed set of algorithms, this work introduces a flexible and general optimization philosophy that combines artificial intelligence, parametric modeling and interactive decision-making. The framework supports informed trade-offs between quantitative criteria and qualitative intent, enabling the generation of efficient, sustainable and context-aware structural solutions. Future developments may extend this approach toward more natural forms of interaction and collaborative design scenarios, as well as toward the integration of interpretability tools to better explain the learned preference models, further strengthening the integration between human expertise and computational optimization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT (OpenAI) to assist with language editing and improving readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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